Learning to Count Anything: Reference-less Class-agnostic Counting with Weak Supervision

Michael A. Hobley Victor A. Prisacariu
Active Vision Lab, University of Oxford
{mahobley, victor}@robots.ox.ac.uk

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Glossary

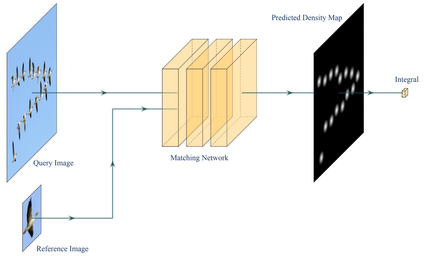
- **Class-agnostic Counting:** Enumerating objects of a novel class.
- **Reference-Less (Exemplar Free):** Counting without the need for an exemplar.
- **Weakly-Supervised:** Training using an integer label instead of a density map.

Standard Method

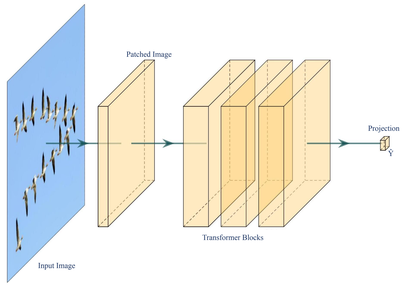
Counting is at its heart object and repetition recognition.

Previous class-agnostic counting methods generally utilise some form of explicit matching between a reference and the query image.

We show that transformers more specifically their use of self-attention can perform this implicitly.



Proposed Method



Paper Highlights:

- A **simple** architecture that achieves **state-of-the-art** class agnostic counting results.
- The **first** method that uses **no reference** images during inference or training.
- The **first** class-agnostic counting method that utilises **only scalar supervision** instead of requiring density maps for training.

Loss

Other methods use density map based supervision:

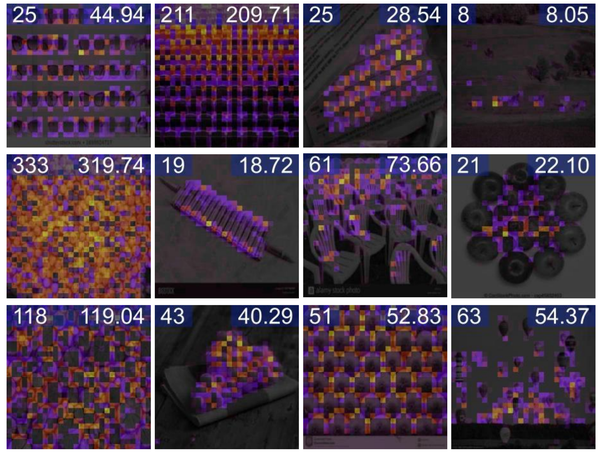
$$\mathcal{L}_{MSE-pix} = \frac{1}{HW}\sum ||y_i - \hat{y}_i||_2^2$$

Our method uses only scalar count values as supervision:

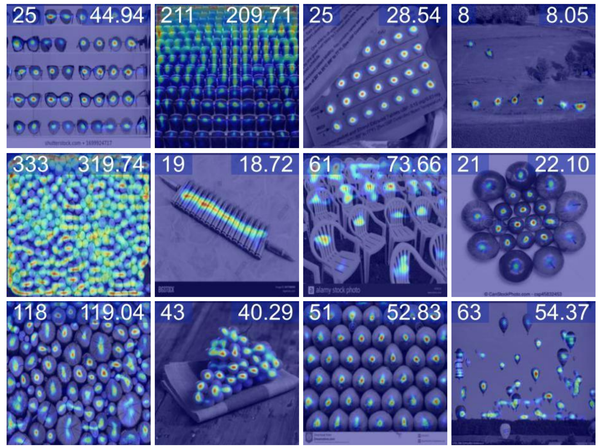
$$\mathcal{L}_{MAPE} = \frac{|\hat{Y} - Y|}{Y}$$

Qualitative Results

Latent Feature Principle Component

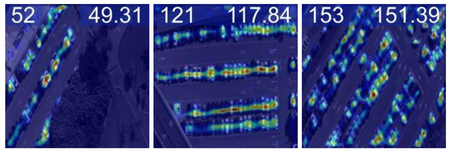


Supervised Localisation Head

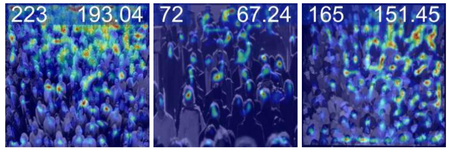


Generalisation

CARPK

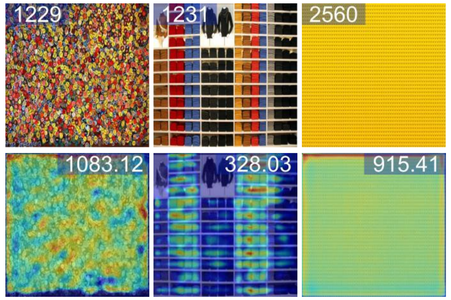


ShanghaiA



Failure Cases and Limitations

High Density



Multi-Class

Quantitative Results

Method			Val MAE	Val RMSE	Test MAE	Test RMSE
Mean			53.38	124.53	47.55	147.67
Median			48.68	129.70	47.73	152.46
Reference-based training						
MAML*	[Finn 2017]	(≥1-shot)	25.54	79.44	24.90	112.68
FR	[Kang 2019]	(≥1-shot)	45.45	112.53	41.64	141.04
FSOD	[Fan 2020]	(≥1-shot)	36.36	115.00	32.53	140.65
GMN	[Lu 2018]	(1-shot)	29.66	89.81	26.52	124.57
FamNet	[Ranjan 2021]	(3-shot)	24.32	70.94	22.56	101.54
FamNet+	[Ranjan 2021]	(3-shot)	23.75	69.07	22.08	99.54
FamNet†	[Ranjan 2021]	(3-shot)	37.90	109.76	33.51	137.79
FamNet+†	[Ranjan 2021]	(3-shot)	37.77	109.04	33.18	137.20
CFOCNet	[Yang 2021]	(3-shot)	21.19	61.41	22.10	112.71
LaoNet	[Lin 2021]	(1-shot)	17.11	56.81	15.78	97.15
BMNet	[Shi 2022]	(3-shot)	19.06	67.95	16.71	103.31
BMNet+	[Shi 2022]	(3-shot)	15.74	58.53	14.62	91.83
BMNet†	[Shi 2022]	(3-shot)	19.29	68.58	18.34	115.31
BMNet+†	[Shi 2022]	(3-shot)	17.21	60.18	16.90	107.66
CounTR	[Liu 2022]	(3-shot)	**13.13**	**49.83**	**11.95**	**91.23**
CounTR†	[Liu 2022]	(3-shot)	22.10	73.08	19.52	111.50
CounTR	[Liu 2022]	(0-shot)	**17.40**	**70.33**	**14.12**	**108.01**
CounTR†	[Liu 2022]	(0-shot)	28.96	90.66	25.58	110.39
Reference-less Training						
MAML*	[Finn 2017]	(0-shot)	32.44	101.08	31.47	129.31
GMN*	[Lu 2018]	(0-shot)	39.02	106.06	37.86	141.39
FamNet*	[Ranjan 2021]	(0-shot)	32.15	98.75	32.27	131.46
RepRPN	[Ranjan 2022]	(0-shot)	29.24	98.11	26.66	129.11
CounTR†*	[Liu 2022]	(0-shot)	33.15	100.80	28.95	125.31
RCC(ours)	[Hobley 2022]	(0-shot)	**17.49**	**58.81**	**17.12**	**104**.53

Code and Paper
countinganything.active.vision